

Creation Date 12-Sep-2014

Revision Date 30-Nov-2024

Revision Number 7

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **tert-Butyl hydroperoxide, 70% aqueous solution**  
Cat No. : **A13926**  
Synonyms TBHP; Trigonox®4 A  
Molecular Formula C<sub>4</sub> H<sub>10</sub> O<sub>2</sub>

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

#### Poison Centre - Emergency information services

**Ireland** : National Poisons Information Centre (NPIC) -  
**01 809 2166** (8am-10pm, 7 days a week)  
**Malta** : +356 2395 2000  
**Cyprus** : +357 2240 5611

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Flammable liquids  
Organic peroxides

Category 3 (H226)  
Type F (H242)

### Health hazards

Acute oral toxicity  
Acute dermal toxicity  
Acute Inhalation Toxicity - Vapors  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Skin Sensitization  
Germ Cell Mutagenicity  
Carcinogenicity

Category 4 (H302)  
Category 3 (H311)  
Category 3 (H331)  
Category 1 C (H314)  
Category 1 (H318)  
Category 1 (H317)  
Category 2 (H341)  
Category 2 (H351)

### Environmental hazards

Chronic aquatic toxicity

Category 2 (H411)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

### **Hazard Statements**

H226 - Flammable liquid and vapor  
H242 - Heating may cause a fire  
H302 - Harmful if swallowed  
H311 + H331 - Toxic in contact with skin or if inhaled  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H341 - Suspected of causing genetic defects  
H351 - Suspected of causing cancer  
H411 - Toxic to aquatic life with long lasting effects

### **Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor/physician

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## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

## Section 3: Composition/information on ingredients

### 3.2. Mixtures

| Component                | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                                                                                                                                    |
|--------------------------|-----------|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tert-Butyl hydroperoxide | 75-91-2   | EEC No. 200-915-7 | 70       | Flam. Liq. 3 (H226)<br>Org. Perox. F (H242)<br>Acute Tox. 4 (H302)<br>Acute Tox. 3 (H311)<br>Acute Tox. 2 (H330)<br>Skin Corr. 1C (H314)<br>Skin Sens. 1 (H317)<br>Eye Dam. 1 (H318)<br>Muta. 2 (H341)<br>Carc. 2 (H351)<br>Aquatic Chronic 2 (H411) |
| Water                    | 7732-18-5 | 231-791-2         | 30       | -                                                                                                                                                                                                                                                    |

| Component                | Specific concentration limits (SCL's)                                          | M-Factor | Component notes |
|--------------------------|--------------------------------------------------------------------------------|----------|-----------------|
| tert-Butyl hydroperoxide | Eye Dam. 1 : C ≥ 1 %<br>Skin Sens. 1 : C ≥ 0.1 %<br>STOT SE 3 : 5 % ≤ C ≤ 10 % | -        | -               |

| Components               | Reach Registration Number |
|--------------------------|---------------------------|
| tert-Butyl hydroperoxide | 01-2119446670-40          |

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to                                                                                                                                                                                                                                       |

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protect themselves and prevent spread of contamination.

## **4.2. Most important symptoms and effects, both acute and delayed**

Causes burns by all exposure routes. Difficulty in breathing. May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## **4.3. Indication of any immediate medical attention and special treatment needed**

### **Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

## **Section 5: Firefighting measures**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Water mist may be used to cool closed containers. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

Do not use halon type extinguisher.

### **5.2. Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.). Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), tert-butanol.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **Section 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### **6.2. Environmental precautions**

Do not flush into surface water or sanitary sewer system.

### **6.3. Methods and material for containment and cleaning up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material. Sweep up and shovel into suitable containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

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## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from clothing and other combustible materials. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep in a dry place. Do not store near combustible materials. Keep away from heat, sparks and flame. Keep refrigerated. To maintain product quality. Do not freeze. Organic peroxides. Corrosives area. Keep at temperature not exceeding 35°C. Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 5.2

**Switzerland - Storage of hazardous substances**

Storage class - SC 5  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s):

| Component                | Latvia                   | Lithuania                         | Luxembourg | Malta | Romania |
|--------------------------|--------------------------|-----------------------------------|------------|-------|---------|
| tert-Butyl hydroperoxide | TWA: 5 mg/m <sup>3</sup> | TWA: 5 mg/m <sup>3</sup> IPRD Oda |            |       |         |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

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## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                  | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|--------------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| tert-Butyl hydroperoxide<br>75-91-2 ( 70 ) |                              |                                 |                                | DNEL = 12.5mg/kg<br>bw/day        |

| Component                                  | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|--------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| tert-Butyl hydroperoxide<br>75-91-2 ( 70 ) | DNEL = 21.34mg/m <sup>3</sup>    | DNEL = 10.37mg/m <sup>3</sup>       | DNEL = 3.69mg/m <sup>3</sup>       | DNEL = 3.08mg/m <sup>3</sup>          |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                  | Fresh water          | Fresh water sediment                  | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)           |
|--------------------------------------------|----------------------|---------------------------------------|--------------------|------------------------------------|------------------------------|
| tert-Butyl hydroperoxide<br>75-91-2 ( 70 ) | PNEC =<br>0.0015mg/L | PNEC =<br>0.00621mg/kg<br>sediment dw | PNEC = 0.015mg/L   | PNEC = 0.17mg/L                    | PNEC =<br>0.166mg/kg soil dw |

| Component                                  | Marine water          | Marine water sediment                  | Marine water Intermittent | Food chain              | Air |
|--------------------------------------------|-----------------------|----------------------------------------|---------------------------|-------------------------|-----|
| tert-Butyl hydroperoxide<br>75-91-2 ( 70 ) | PNEC =<br>0.00015mg/L | PNEC =<br>0.000621mg/kg<br>sediment dw |                           | PNEC = 1.4mg/kg<br>food |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                    | Glove thickness | EU standard | Glove comments        |
|----------------|--------------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers<br>recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

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Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

## Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                                                    |                          |                          |
|----------------------------------------------------|--------------------------|--------------------------|
| Physical State                                     | Liquid                   |                          |
| Appearance                                         | Colorless                |                          |
| Odor                                               | Slight                   |                          |
| Odor Threshold                                     | No data available        |                          |
| Melting Point/Range                                | -2.8 °C / 27 °F          |                          |
| Softening Point                                    | No data available        |                          |
| Boiling Point/Range                                | 37 °C / 98.6 °F          | @ 15 mmHg                |
| Flammability (liquid)                              | Flammable                | On basis of test data    |
| Flammability (solid,gas)                           | Not applicable           | Liquid                   |
| Explosion Limits                                   | No data available        |                          |
| Flash Point                                        | 43 °C / 109.4 °F         | Method - CC (closed cup) |
| Autoignition Temperature                           | 204 °C / 399.2 °F        |                          |
| Decomposition Temperature                          | 75 °C                    |                          |
| Self-Accelerating Decomposition Temperature (SADT) | 80°C                     |                          |
| pH                                                 | 4.3 @ 20°C               | sat.aq.sol               |
| Viscosity                                          | 4.1 mPa.s at 20 °C       |                          |
| Water Solubility                                   | Miscible                 |                          |
| Solubility in other solvents                       | No information available |                          |
| Partition Coefficient (n-octanol/water)            |                          |                          |
| Component                                          | log Pow                  |                          |
| tert-Butyl hydroperoxide                           | 0.846                    |                          |
| Vapor Pressure                                     | 62 mmHg @ 45 °C          |                          |
| Density / Specific Gravity                         | 0.940                    |                          |
| Bulk Density                                       | Not applicable           | Liquid                   |
| Vapor Density                                      | 3.1 (Air = 1.0)          | (Air = 1.0)              |
| Particle characteristics                           | Not applicable (liquid)  |                          |

### 9.2. Other information

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Molecular Formula C4 H10 O2  
Molecular Weight 90.12  
Explosive Properties explosive air/vapour mixtures possible  
Oxidizing Properties Oxidizer

## Section 10: Stability and reactivity

### 10.1. Reactivity

Yes

### 10.2. Chemical stability

Oxidizer: Contact with combustible/organic material may cause fire.

### 10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.  
Hazardous Reactions None under normal processing.

### 10.4. Conditions to avoid

Do not freeze. Temperatures above 35°C. Keep away from open flames, hot surfaces and sources of ignition. Exposure to light. Incompatible products. Combustible material. Excess heat.

### 10.5. Incompatible materials

Organic materials. Acids. Bases. Metals. Reducing Agent. Finely powdered metals. Strong reducing agents. Combustible material.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). tert-butanol.

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral Category 4  
ATE = 800 mg/kg  
Dermal Category 3  
ATE = 897 mg/kg  
Inhalation Category 3  
ATE = 2.64 mg/l

#### Toxicology data for the components

| Component                | LD50 Oral                | LD50 Dermal                 | LC50 Inhalation                           |
|--------------------------|--------------------------|-----------------------------|-------------------------------------------|
| tert-Butyl hydroperoxide | LD50 = 560 mg/kg ( Rat ) | LD50 = 628 mg/kg ( Rabbit ) | LC50 = 1845 mg/m <sup>3</sup> ( Rat ) 4 h |
| Water                    | -                        | -                           | -                                         |

(b) skin corrosion/irritation; Category 1 C

(c) serious eye damage/irritation; Category 1



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**(d) respiratory or skin sensitization;**

**Respiratory**

No data available

**Skin**

Category 1

May cause sensitization by skin contact

**(e) germ cell mutagenicity;**

Category 2

Possible risk of irreversible effects

**(f) carcinogenicity;**

Category 2

**(g) reproductive toxicity;**

No data available

**(h) STOT-single exposure;**

No data available

**(i) STOT-repeated exposure;**

No data available

**Target Organs**

None known.

**(j) aspiration hazard;**

No data available

**Symptoms / effects, both acute and delayed**

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties**

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## Section 12: Ecological information

### 12.1. Toxicity

**Ecotoxicity effects**

The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component                | Freshwater Fish                                                                                                      | Water Flea                              | Freshwater Algae                                           |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------------------|
| tert-Butyl hydroperoxide | LC50: = 42.3 mg/L, 96h<br>semi-static (Pimephales<br>promelas)<br>LC50: = 57 mg/L, 96h static<br>(Brachydanio rerio) | EC50: = 20 mg/L, 48h (Daphnia<br>magna) | EC50: = 2.1 mg/L, 72h<br>(Pseudokirchneriella subcapitata) |

### 12.2. Persistence and degradability

**Persistence**

Not readily biodegradable

**Degradation in sewage**

Persistence is unlikely, based on information available.

Contains substances known to be hazardous to the environment or not degradable in waste

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treatment plant water treatment plants.

**12.3. Bioaccumulative potential** Bioaccumulation is unlikely

| Component                | log Pow | Bioconcentration factor (BCF) |
|--------------------------|---------|-------------------------------|
| tert-Butyl hydroperoxide | 0.846   | No data available             |

**12.4. Mobility in soil** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air

**12.5. Results of PBT and vPvB assessment** No data available for assessment.

**12.6. Endocrine disrupting properties**  
**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**12.7. Other adverse effects**  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13: Disposal considerations

### **13.1. Waste treatment methods**

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information** Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment.

**Switzerland - Waste Ordinance** Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## Section 14: Transport information

### **IMDG/IMO**

**14.1. UN number** UN3109  
**14.2. UN proper shipping name** ORGANIC PEROXIDE TYPE F, LIQUID  
**Technical Shipping Name** tert-Butyl hydroperoxide  
**14.3. Transport hazard class(es)** 5.2  
**Subsidiary Hazard Class** 8  
**14.4. Packing group**

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## ADR

|                                         |                                 |
|-----------------------------------------|---------------------------------|
| <b>14.1. UN number</b>                  | UN3109                          |
| <b>14.2. UN proper shipping name</b>    | ORGANIC PEROXIDE TYPE F, LIQUID |
| <b>Technical Shipping Name</b>          | tert-Butyl hydroperoxide        |
| <b>14.3. Transport hazard class(es)</b> | 5.2                             |
| <b>Subsidiary Hazard Class</b>          | 8                               |
| <b>14.4. Packing group</b>              |                                 |

## IATA

|                                         |                                 |
|-----------------------------------------|---------------------------------|
| <b>14.1. UN number</b>                  | UN3109                          |
| <b>14.2. UN proper shipping name</b>    | ORGANIC PEROXIDE TYPE F, LIQUID |
| <b>Technical Shipping Name</b>          | tert-Butyl hydroperoxide        |
| <b>14.3. Transport hazard class(es)</b> | 5.2                             |
| <b>Subsidiary Hazard Class</b>          | 8                               |
| <b>14.4. Packing group</b>              |                                 |

|                                    |                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>14.5. Environmental hazards</b> | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
|------------------------------------|----------------------------------------------------------------------------------------------------------|

|                                           |                                  |
|-------------------------------------------|----------------------------------|
| <b>14.6. Special precautions for user</b> | No special precautions required. |
|-------------------------------------------|----------------------------------|

|                                                                      |                                |
|----------------------------------------------------------------------|--------------------------------|
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods |
|----------------------------------------------------------------------|--------------------------------|

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

X = listed. US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| tert-Butyl hydroperoxide | 75-91-2   | 200-915-7 | -      | -   | X     | X    | KE-11387 | X    | X    |
| Water                    | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |

| Component                | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| tert-Butyl hydroperoxide | 75-91-2   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water                    | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component                | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| tert-Butyl hydroperoxide | 75-91-2   | -                                                                   | Use restricted. See entry 75. (see link for restriction details)              | -                                                                                                     |
| Water                    | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |

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## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component                | CAS No    | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Major Accident<br>Notification | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Safety Report<br>Requirements |
|--------------------------|-----------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| tert-Butyl hydroperoxide | 75-91-2   | Not applicable                                                                                  | Not applicable                                                                                 |
| Water                    | 7732-18-5 | Not applicable                                                                                  | Not applicable                                                                                 |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 3 (self classification)

| Component                | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|--------------------------|---------------------------------------|-------------------------|
| tert-Butyl hydroperoxide | WGK3                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## Section 16: Other information

### Full text of H-Statements referred to under sections 2 and 3

H242 - Heating may cause a fire  
H302 - Harmful if swallowed  
H311 - Toxic in contact with skin  
H331 - Toxic if inhaled  
H314 - Causes severe skin burns and eye damage  
H317 - May cause an allergic skin reaction  
H318 - Causes serious eye damage  
H341 - Suspected of causing genetic defects  
H351 - Suspected of causing cancer  
H411 - Toxic to aquatic life with long lasting effects  
H226 - Flammable liquid and vapor  
H330 - Fatal if inhaled

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## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### Training Advice

Chemical incident response training.

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

**Prepared By** Health, Safety and Environmental Department

**Creation Date** 12-Sep-2014

**Revision Date** 30-Nov-2024

**Revision Summary** SDS sections updated.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other

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materials or in any process, unless specified in the text

**End of Safety Data Sheet**